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AUTOMATED PLANT DISEASE DETECTION USING CONVOLUTIONAL NEURAL NETWORKS

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AUTOMATED PLANT DISEASE DETECTION USING CONVOLUTIONAL NEURAL NETWORKS

INTRODUCTION

- **Agriculture Is The Backbone Of Food Security, But Plant Diseases Severely Affect Crop Yield And Quality.**

- **Traditional Methods Of Detecting Diseases Rely On Human Experts, Which Is Slow, Inconsistent, And Not Available In Remote Areas.**

- **Convolutional Neural Networks (Cnns) Can Automatically Analyze Leaf Images And Identify Diseases With High Accuracy.**

- **Our Project Aims To Create A System That Connects Ai Technology With Agriculture, Helping Farmers Detect Diseases Early And Improve Productivity**

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PROBLEM STATEMENT

- **Plant diseases are a major challenge, causing large-scale losses in crops and farmer income.**

- **Farmers often depend on manual inspection, which is error-prone and delayed.**

- **Lack of early detection allows diseases to spread quickly, leading to bigger damage.**

- **Misdiagnosis also results in wrong pesticide usage, harming both crops and the environment. Hence, there is a need for an accurate, automated, and scalable plant disease detection system.**

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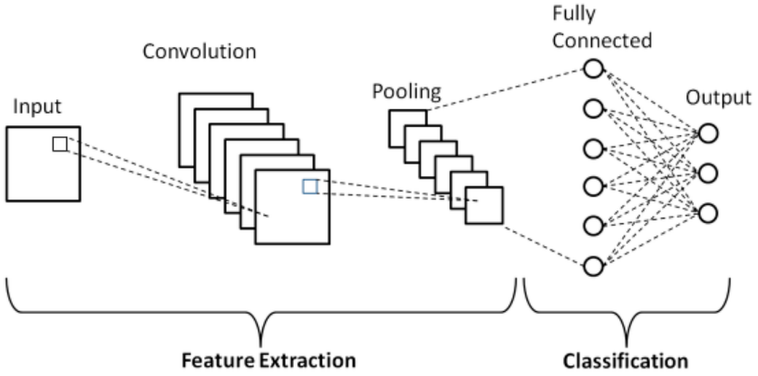
EXISTING SYSTEM

- **Current systems mainly use public datasets like PlantVillage, which contain clean lab images.**

- **The pipeline usually follows steps:**

image collection → CNN model training → classification.

→



- **Models such as VGG16, ResNet, and MobileNet are widely used often with transfer learning.**

- **These systems give a mid level accuracy in controlled datasets, but performance drops in real farm conditions with lighting, clutter, and multiple leaves.**

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PROPOSED SYSTEM

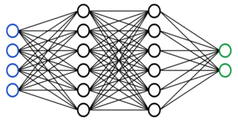
- **Our proposed system is a web-based application developed using TensorFlow and Streamlit.**
- **It uses a large dataset of over 87,000 images across 38 disease classes covering major crops.**
- **A custom CNN model was trained, reaching 99% training accuracy and 97% validation accuracy.**
- **Farmers can simply upload a leaf image, and the app predicts the disease in real-time (under 5 seconds).**
- **The system is cost-effective, accessible, and designed to work even for farmers in rural areas.**

TensorFlow

OpenCV

Streamlit

Keras



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OBJECTIVES

- **Improve diagnosis accuracy and reliability compared to manual inspection.**
- **Help farmers take quick decisions and apply treatments at an early stage.**
- **Build a scalable model that works for different crops and disease types.**
- **Provide a user-friendly web app with real-time prediction capability.**
- **Reduce dependence on lab tests, making it affordable for small-scale farmers.**

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Methodology

1. **Dataset: Collected 87,000+ images, split into training (80%), validation (20%), and test sets.**

2. **Preprocessing: All images resized to 128x128 pixels, normalized, and batched for training.**

3. **Model: Designed a CNN with convolutional, pooling, and dense layers, ReLU for hidden layers and Softmax for output.**

4. **Training: Model trained using keras and Adam optimizer, categorical cross-entropy loss, for 10 epochs.**

5. **Evaluation: Achieving ~99.00% training accuracy and ~97.20% validation accuracy.**

6. **Deployment: Implemented as a Streamlit web app, allowing farmers to upload images and receive predictions instantly.**

Training Accuracy	99%
Validation Accuracy	97%
Prediction Time	<5 seconds

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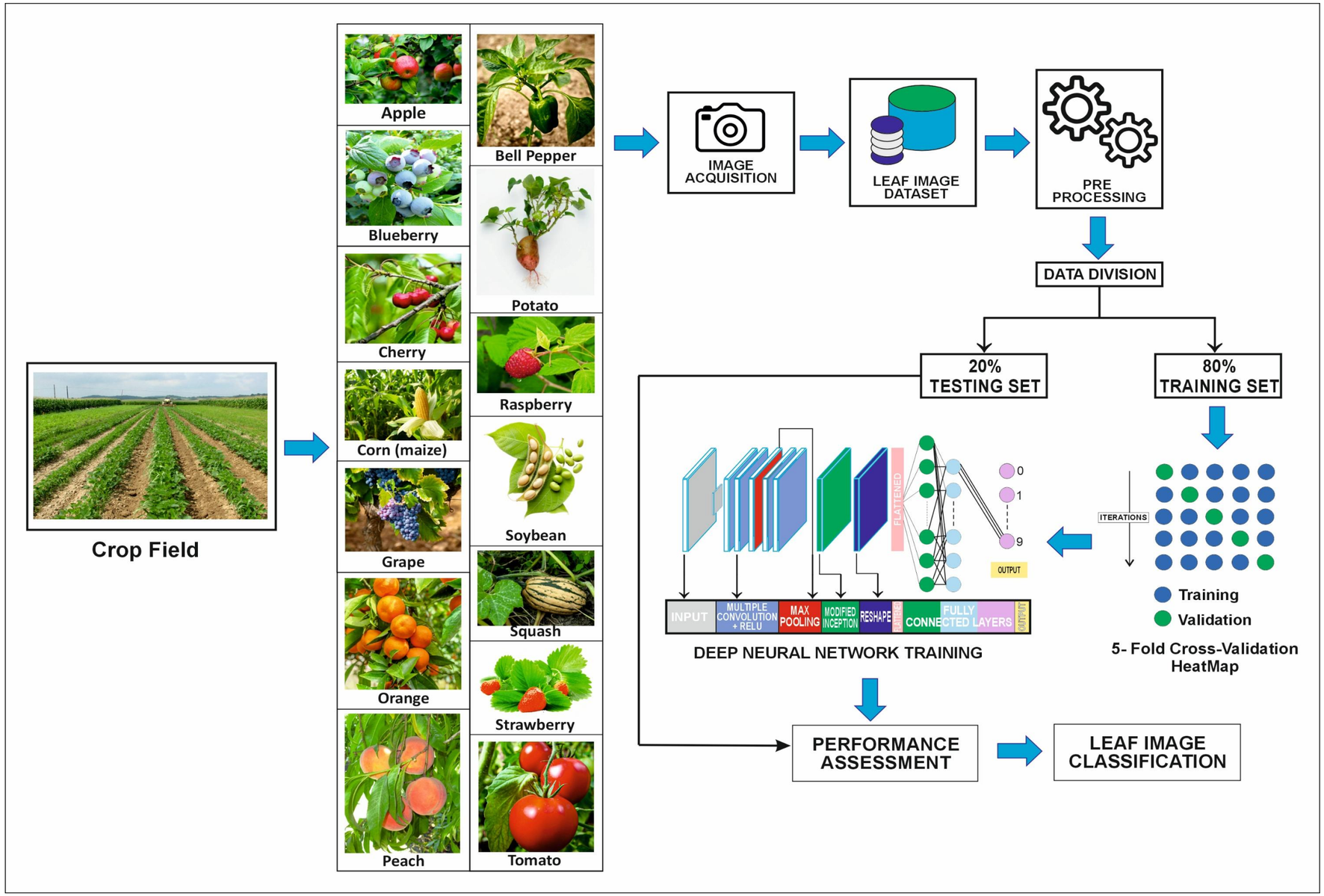
1. **Dataset:**

✓ **Collected 87,000+ images.**

✓ **Split into**

- **training (80%)**
- **test sets & validation (20%)**



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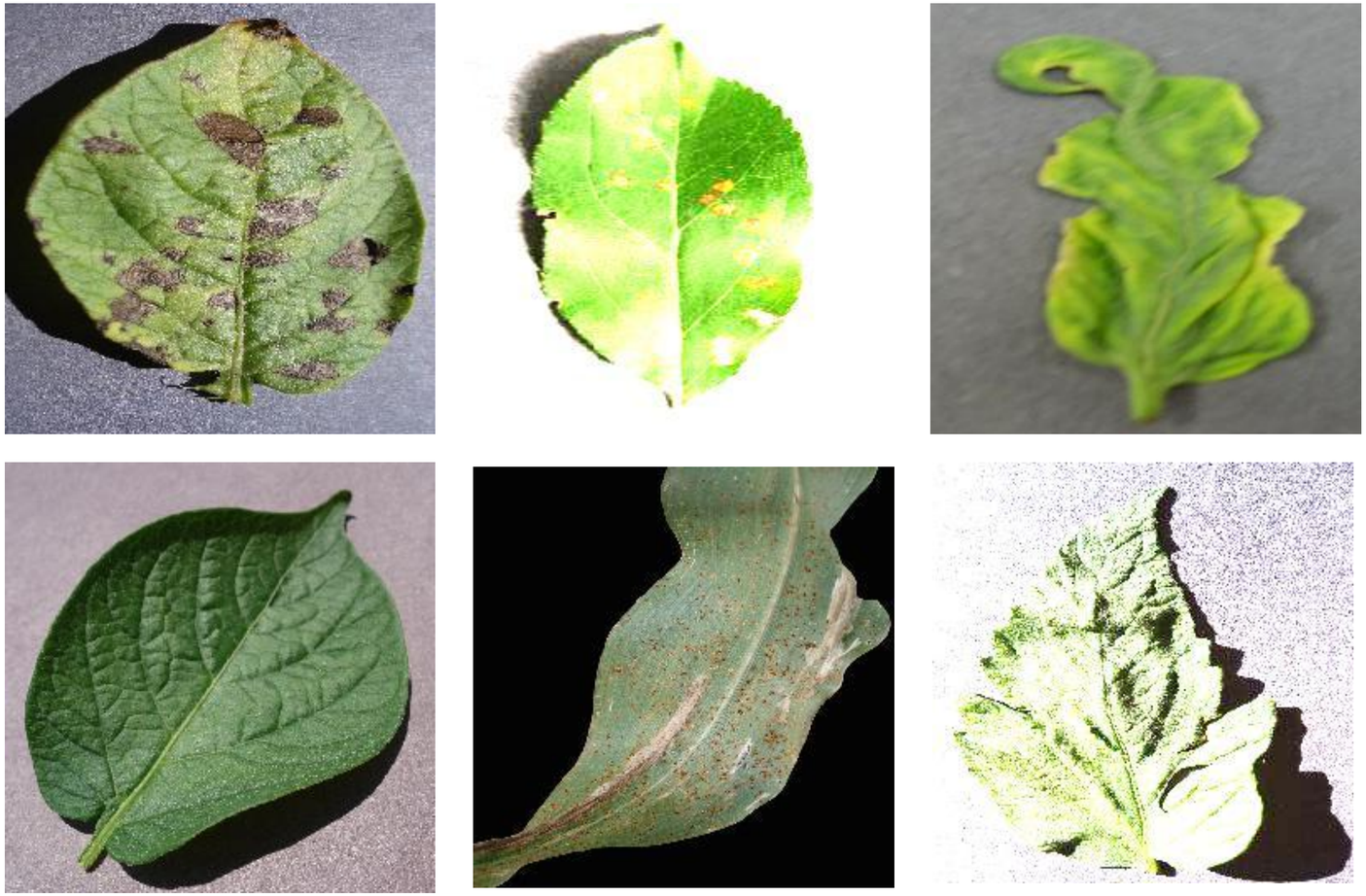
2. Preprocessing:

✓ **All images resized to**

- **128x128 pixels**
- **Normalized**
- **Batched for training.**



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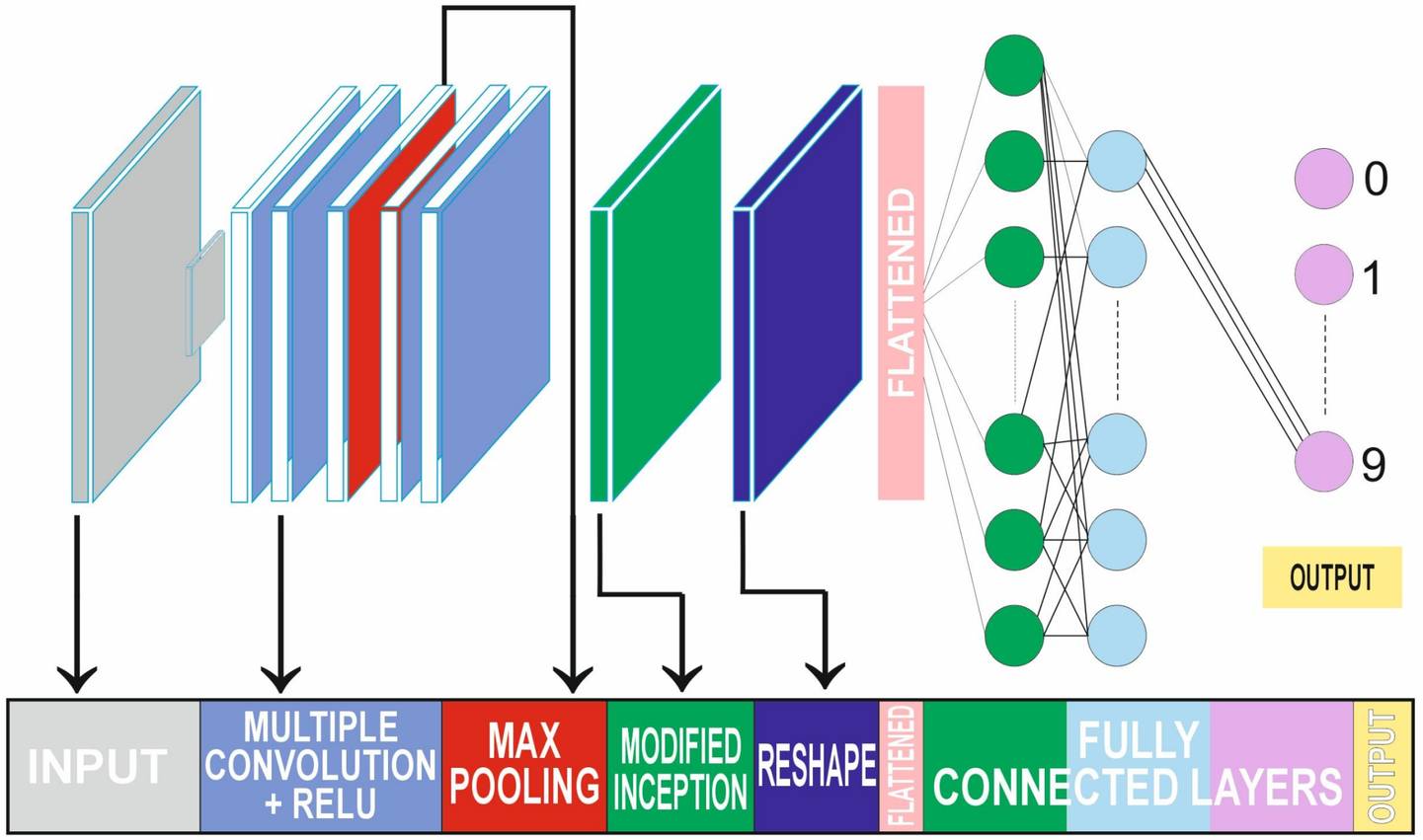
3. Model:

Designed a CNN with

- **convolutional**
- **pooling**
- **dense layers**

  ReLU for hidden layers

  and Softmax for output.



WORKING OF CONVOLUTION NEURAL NETWORK

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4. Training: Model trained using keras and categorical cross- entropy loss for 10 epochs
5. Evaluation: Achieving ~99.00% training accuracy and ~97.20% validation accuracy.

```
Epoch 1/10
2197/2197 [==============================] - 735s 292ms/step - loss: 1.2808 - accuracy: 0.6232 - val_loss: 0.5307 - val_accuracy: 0.8309
Epoch 2/10
2197/2197 [==============================] - 718s 326ms/step - loss: 0.4186 - accuracy: 0.8663 - val_loss: 0.2737 - val_accuracy: 0.9130
Epoch 3/10
2197/2197 [==============================] - 772s 351ms/step - loss: 0.2574 - accuracy: 0.9174 - val_loss: 0.2459 - val_accuracy: 0.9210
Epoch 4/10
2197/2197 [==============================] - 801s 364ms/step - loss: 0.1819 - accuracy: 0.9410 - val_loss: 0.1529 - val_accuracy: 0.9500
Epoch 5/10
2197/2197 [==============================] - 787s 358ms/step - loss: 0.1353 - accuracy: 0.9554 - val_loss: 0.1235 - val_accuracy: 0.9591
Epoch 6/10
2197/2197 [==============================] - 744s 338ms/step - loss: 0.1046 - accuracy: 0.9651 - val_loss: 0.1236 - val_accuracy: 0.9616
Epoch 7/10
2197/2197 [==============================] - 686s 312ms/step - loss: 0.0904 - accuracy: 0.9705 - val_loss: 0.1162 - val_accuracy: 0.9633
Epoch 8/10
2197/2197 [==============================] - 659s 300ms/step - loss: 0.0731 - accuracy: 0.9763 - val_loss: 0.1451 - val_accuracy: 0.9589
Epoch 9/10
2197/2197 [==============================] - 615s 280ms/step - loss: 0.0631 - accuracy: 0.9797 - val_loss: 0.0976 - val_accuracy: 0.9713
Epoch 10/10
2197/2197 [==============================] - 611s 278ms/step - loss: 0.0569 - accuracy: 0.9811 - val_loss: 0.0964 - val_accuracy: 0.9720
```

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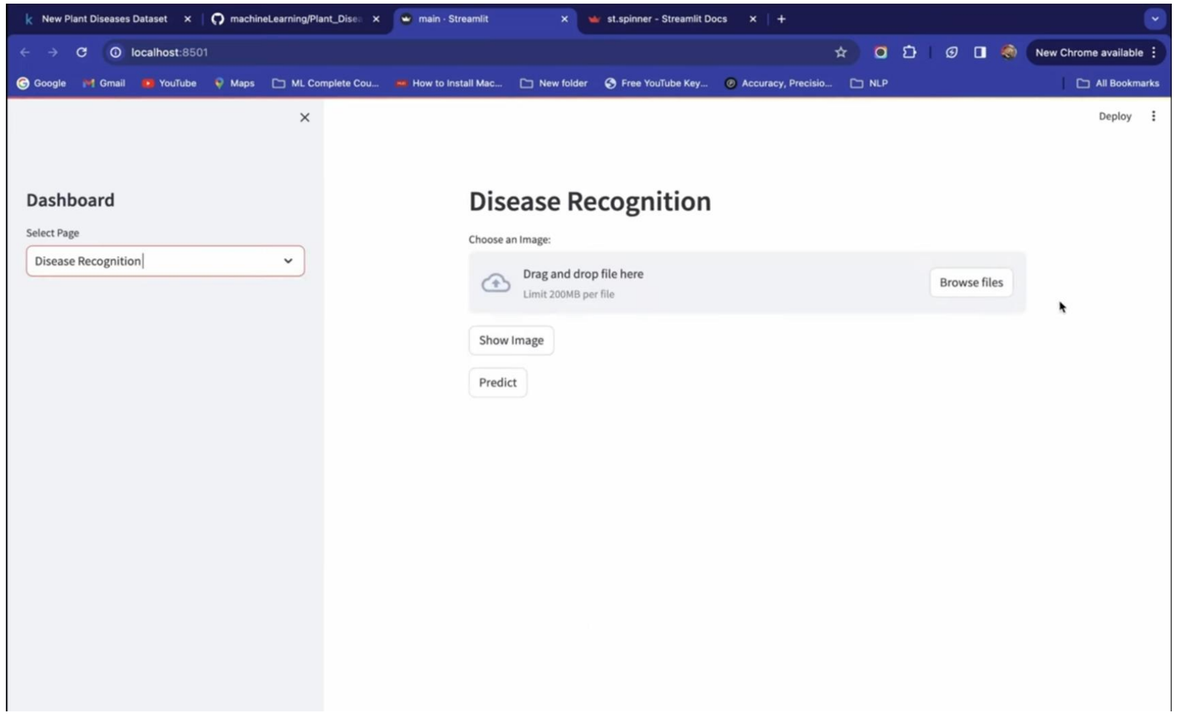
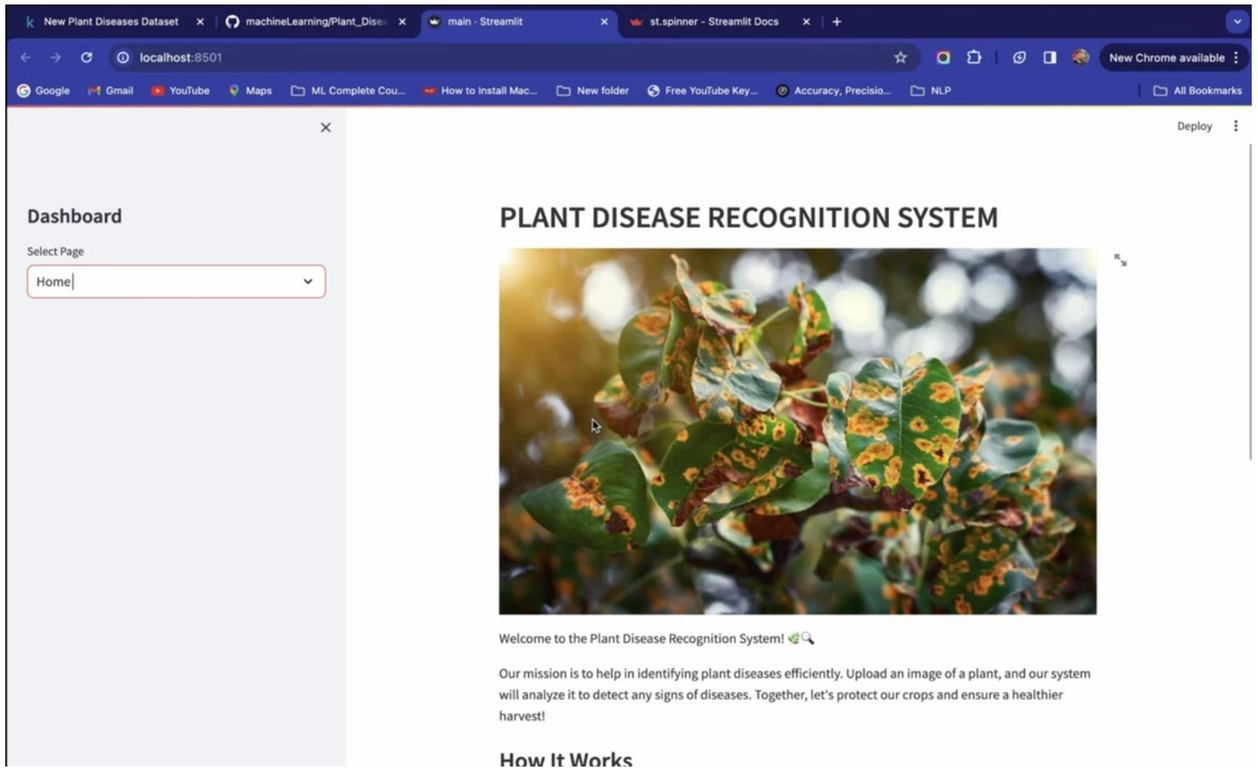
6. Deployment:

- **Implemented as a Streamlit web app, allowing farmers to upload images and receive predictions instantly.**





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Expected Outcomes

- **The system will be able to detect 38 different plant diseases with high accuracy.**
- **Predictions will be generated in real-time, within a <5 seconds of uploading an image.**
- **Early and accurate detection will help reduce crop losses and improve yield.**
- **Farmers will be able to save costs by avoiding unnecessary pesticide usage.**

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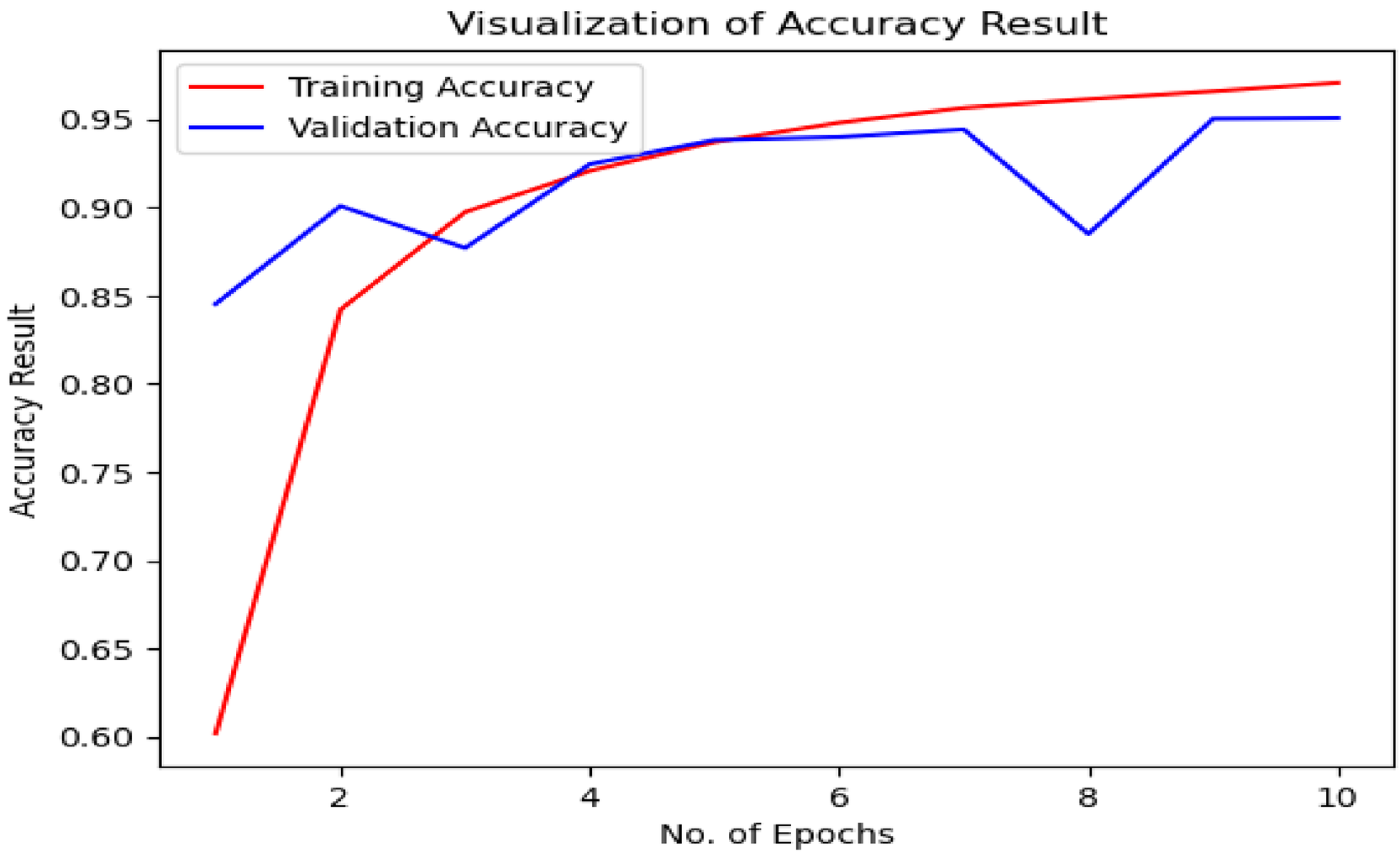
Visualization of Accuracy Result



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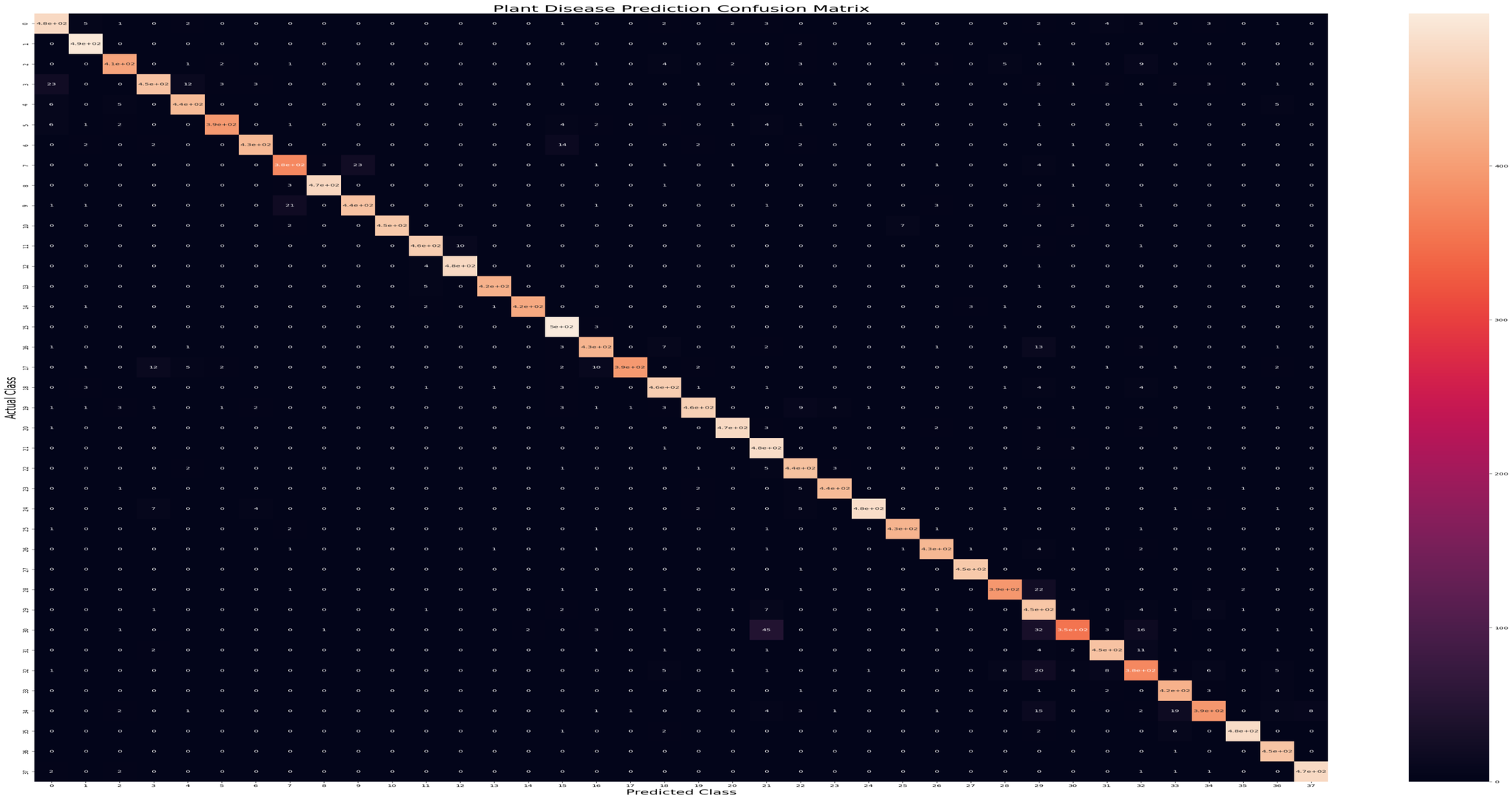
Plant Disease Prediction Confusion Matrix



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SYSTEM REQUIREMENTS

Hardware Requirements

Category	Specification
Minimum Processor	Intel Core i3 (7th Gen) / AMD Ryzen 3
Recommended Processor	Intel Core i5/i7 / AMD Ryzen 5/7
Minimum RAM	8 GB
Recommended RAM	16 GB or higher
Storage	50 GB free space (SSD recommended)
GPU (Minimum)	Integrated Graphics
GPU (Recommended)	NVIDIA GTX 1650/1660 or RTX 2060/3060/4060

Software Requirements

Category	Specification
Operating System	Windows 10/11
Programming Language	Python 3.10
Development Tools	Anaconda, Jupyter Notebook/JupyterLab, VS Code
ML Frameworks	TensorFlow 2.x / Keras OR PyTorch 1.12+
Image Processing Tools	OpenCV, scikit-image
Supporting Tools	Git, MS PowerPoint, MS Word
Optional Deployment Tools	Flask, Streamlit, TensorFlow Lite

Required Python Libraries

Library Category	Libraries Used
Deep Learning	TensorFlow 2.x, Keras, PyTorch (optional), Torchvision
Image Processing	OpenCV-Python, Pillow, scikit-image
Data Handling	NumPy, Pandas
Visualization	Matplotlib, Seaborn
ML / Preprocessing	Scikit-learn, Albumentations, imgaug

Additional Requirements

Category	Specification
Dataset	PlantVillage dataset or custom field dataset
GPU Support	CUDA Toolkit 11.x, cuDNN 8.x (if using NVIDIA GPU)
Documentation Tools	MS Word, Google Docs
Presentation Tools	PowerPoint / Google Slides

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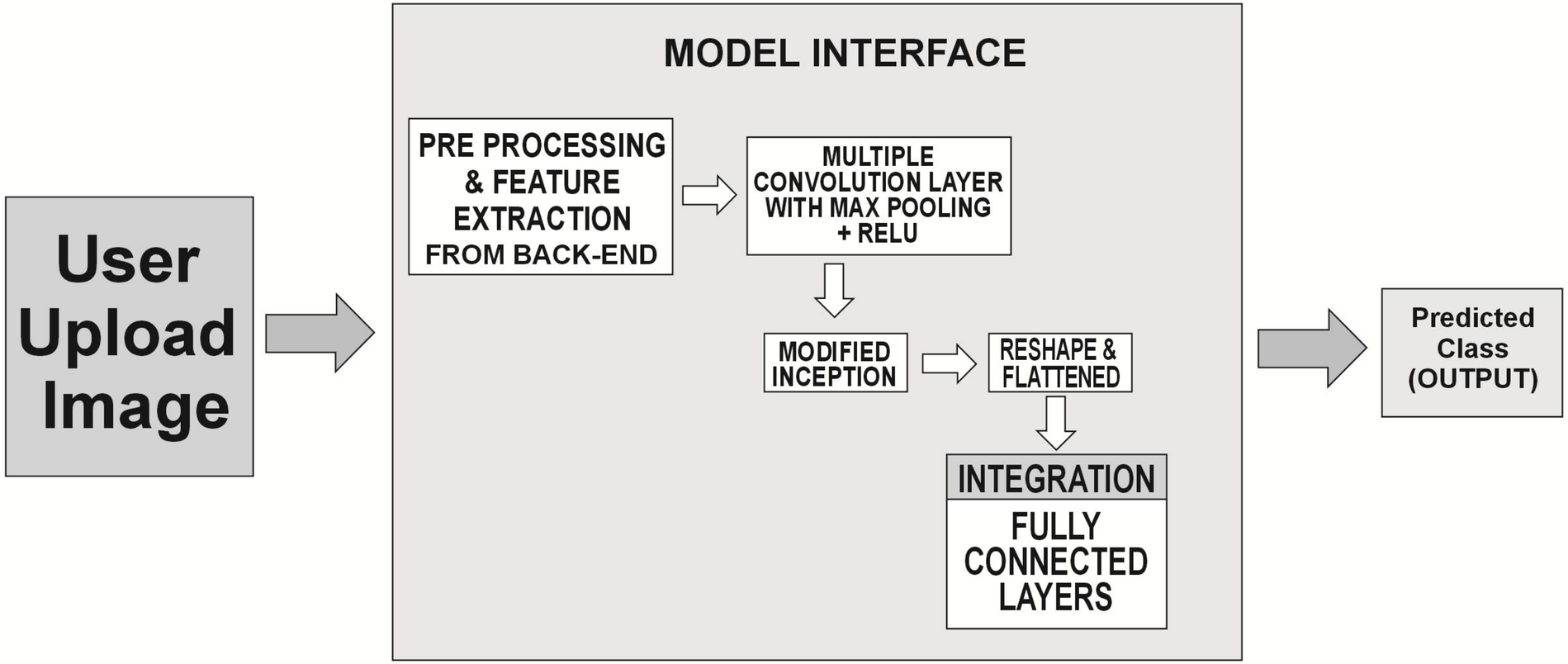
User Work Flow Diagram



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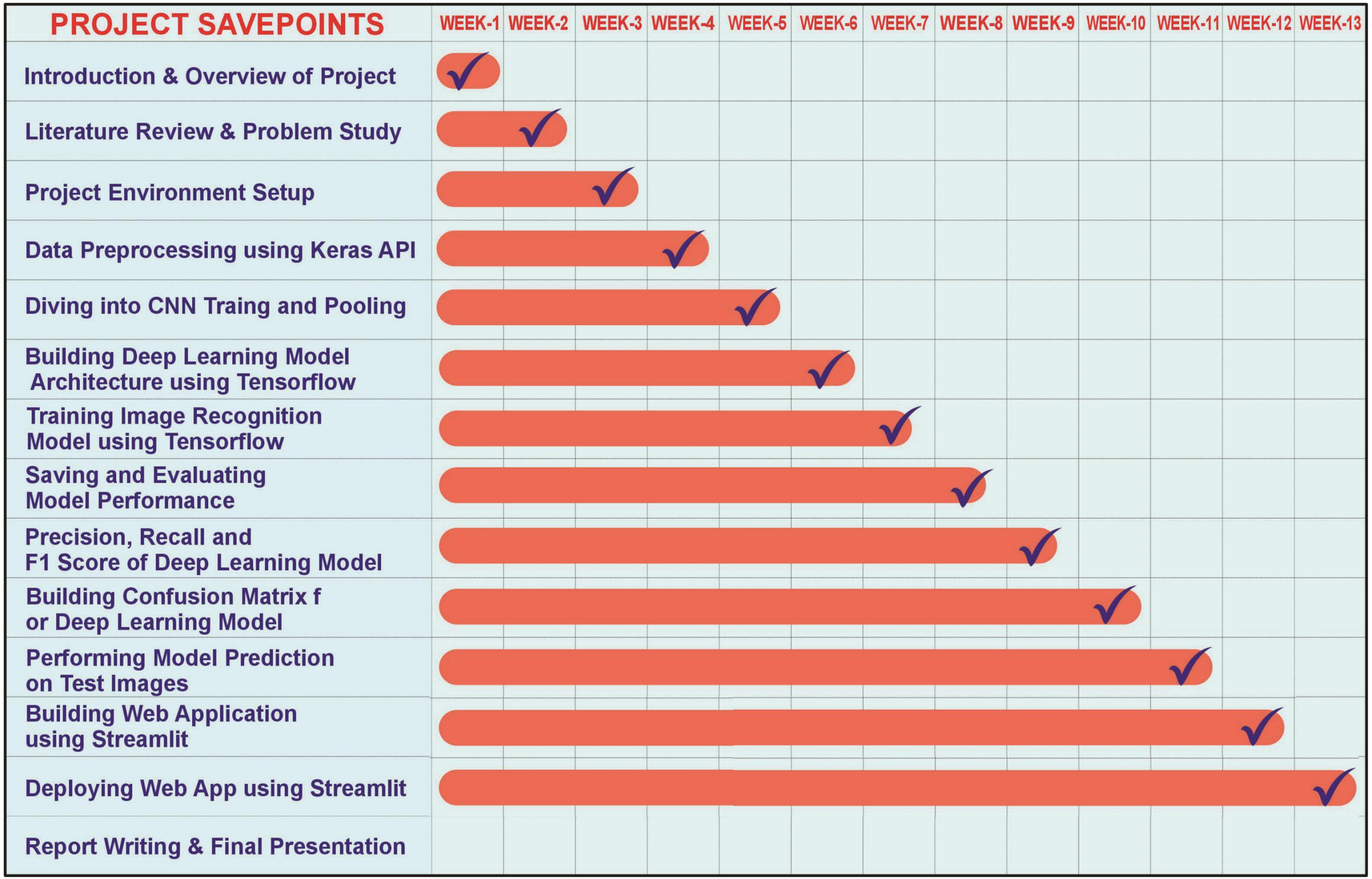
Project Timeline



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Refferences

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Thank you very much!